



Objectives:

1. Proceed with the experimental/simulation work and data collection.
2. Able to design and conduct the experiment/simulation.
3. Choose the appropriate technique to achieve the experiment/simulation goal with proper justification.

Student's Name/Signature:

NAVINNESHSAAN

Date: 17 APRIL 2026

Supervisor's Name/Signature:

Date:



WEEK 1 - WEEK 5

Summary of weekly activities

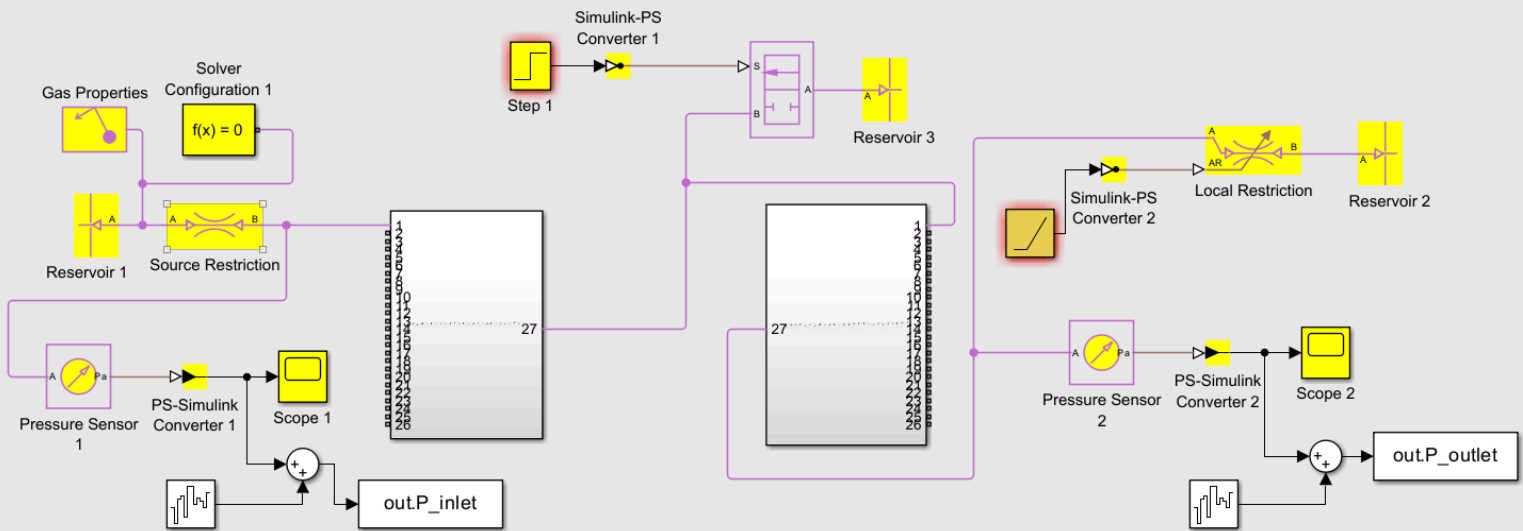
During the first five weeks, the primary focus was optimizing the pipeline simulation model in MATLAB R2025a and Simscape to resolve the numerical stiffness and wave delay issues identified in FYP Part 1. The model was restructured using a dynamic "accordion" method, expanding the pipeline resolution to 50 discrete segments to ensure physically accurate Negative Pressure Wave (NPW) propagation speeds. To accurately represent real-world SCADA systems and generate robust Class 0 (normal operation) data, Band-Limited White Noise blocks were integrated to simulate baseline sensor static. Additionally, a Variable Local Restriction driven by a Ramp block was implemented to model gradual operational transients, such as routine downstream valve adjustments.

Following the physical model optimization, an automated MATLAB script was developed and executed to conduct 200 continuous simulations, comprising 100 Class 1 (leak) scenarios and 100 Class 0 (normal) scenarios. The script dynamically randomized leak locations, rupture sizes, and valve ramp slopes to create a highly varied synthetic dataset. Furthermore, a digital moving average filter and gradient-based (dP/dt) detection logic were coded to automatically extract the precise pressure drop times (t_A and t_B) and calculate the theoretical leak location (X_{calc}). This data extraction process has been successfully completed and compiled into a structured Excel spreadsheet, marking the conclusion of Phase 2 and establishing a fully pre-processed foundation to begin Phase 3: designing and training the Recurrent Neural Network (RNN).

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Summary of weekly activities



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run_fyp_simulation.m * x
C:\Users\ASUS\Downloads\FYP\run_fyp_simulation.m
71 % --- EXTRACT SENSOR DATA ---
72 time = simOut.P_inlet.Time;
73 P_in = simOut.P_inlet.Data;
74 P_out = simOut.P_outlet.Data;
75
76 % --- 1. SIGNAL FILTERING ---
77 P_in_clean = smoothdata(P_in, 'movmean', 50);
78 P_out_clean = smoothdata(P_out, 'movmean', 50);
79
80 % --- 2. GRADIENT LOGIC (FIND THE DROP) ---
81 dt_sample = mean(diff(time)); % Calculate the average time step
82 dP_in_dt = diff(P_in_clean) ./ dt_sample;
83 dP_out_dt = diff(P_out_clean) ./ dt_sample;
84
85 gradient_threshold = -0.02;
86
87 index_A = find(dP_in_dt < gradient_threshold, 1);
88 index_B = find(dP_out_dt < gradient_threshold, 1);
89
90 % --- 3. RECORD TIMES AND CALCULATE LOCATION ---
91 if ~isempty(index_A)
92     t_A(i) = time(index_A);
93 else
94     t_A(i) = NaN; % Logs as NaN in Excel if no leak detected
95 end
96
97 if ~isempty(index_B)
98     t_B(i) = time(index_B);
99 else
100    t_B(i) = NaN; % Logs as NaN in Excel if no leak detected
101 end
102
103 % Calculate X_calc using the NPW formula
104 if ~isnan(t_A(i)) && ~isnan(t_B(i))
105     delta_t(i) = t_A(i) - t_B(i);

```

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